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The Effects of Udder Edema on Parlor Behavior in First and Second Lactation Holstein Dairy Cows

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ABSTRACT

Udder edema (UE) is a metabolic disorder that most commonly occurs around the transition period. However, there are gaps in our knowledge about its effects on parlor behavior. The aim of this study was to examine the prevalence and effects UE has on behavior in the milking parlor during udder preparation and active milking from 2 to 9 DIM in first (FL) and second lactation (SL) Holstein dairy cattle. Each cow ($n = 375$) was observed once and monitored from the point of first contact during udder preparation through the first 5 min of active milking. Behavior measurements include step, kick, and kicking off the milking unit (KOU). Cows were determined to have UE when the rear udder medial suspensory ligament lost definition and was softened due to the presence of interstitial fluid. In this study, 237/247 (95.6%) FL cows and 104/128 (81.3%) SL cows presented with UE. First lactation cows with UE had a higher step rate (3.97 steps/session) when in contact during udder preparation and attachment (2.80 steps/session), and kick rate during milking (2.68 kicks/milking session) when compared with SL cows with UE (2.37 steps/session, 1.25 steps/session, 1.24 kicks/milking session). FL cows with UE had a lower step rate during milking (6.04 steps/milking session) compared with FL cows without UE (7.20 steps/milking session). FL cows with UE had a higher average KOU count ($\mu = 0.220$ kick-offs) than SL cows with UE ($\mu = 0.029$ kick-offs) and FL cows without UE ($\mu = 0.091$ kick-offs). The results indicate that UE is prevalent among dairy cattle, and the disorder has effects on behaviors presented in the milking parlor. Reducing UE has the potential to decrease step and kick behavior to improve welfare of transition cows and reduce risk to dairy caretakers in the milking parlor.

Key words: udder edema, udder oedema, dairy cattle, behavior

INTRODUCTION

Udder edema (UE) is a metabolic disorder where lymphatic fluid is trapped in the tissues of the udder around the time of parturition (Bacic et al., 2007). Documented risk factors for peri- and post-parturient UE include physiological changes during a heifer's gestational period (Okkema and Grandin, 2021), age at first calving (Dentine and McDaniel, 1983), genetic predisposition (Norman et al., 1974, Van Dorp et al., 1998), nutrition (Randall et al., 1974, Moore et al., 2000), and oxidative stress (Miller et al., 1993). Udder edema negatively impacts milk production (Melendez et al., 2006), increases the risk for udder and teat injuries, chronic mastitis (Bacic et al., 2007), udder cleft dermatitis (Beattie and Taylor, 2000), and permanently damages mammary tissues (Dentine and McDaniel, 1984). Melendez et al. (2006) theorized that UE causes discomfort to cows and impacts parlor efficiency. It has been documented that edematous udders are more difficult to attach a milking unit because of swollen and malformed teats (Melendez et al., 2006). When examining first lactation cows, edematous udders are a common occurrence in United States dairy cattle, at 97% (Dentine and McDaniel, 1983). Morrison et al. (2018) had similar findings on Canadian dairy farms, where 86% of first parity cows had UE; In contrast, Erb and Grohn (1988), found a lactational incidence rate of UE ranging from 1.1 to 18% in Finnish dairy herds. Cows who experience UE in their first lactation are at greater risk for its recurrence in future lactations (Melendez et al., 2006). Udder edema is shown to have a negative impact on the health and productive ability of dairy cows and poses a risk for future complications, but it is not known if UE affects a cow's behavior and subsequently their welfare while in the milking parlor.

Physiological, pathological, and injection-induced bovine udder edema have differing presentations (Waller, O'Brien, and McGuirk, 2007). Udder edema develops under 3 different circumstances: as a result of over-bagging, injection induced, and the transition period, which is the most common and the focus of this study. Visual assessment of postpartum UE has been used in

various scoring tools, ranging from 4-point to 10-point scales (Dentine and McDaniel, 1983, Nestor et al., 1987, Tucker et al., 1992, Kojouri et al., 2015, Morrison et al., 2018). For scores with adequate descriptions, the beginning stages of udder edema include swelling at the base of the udder, causing uneven quarters and the appearance of the medial cleft ligament to be reduced. No industry-wide standardized method for visually scoring peri- and postpartum UE has been developed at this time.

The overall methodology of the study was to examine UE prevalence and milking parlor behavior with a farm manager's mindset. Physiological applications of pain assessment can be sensitive to stress and may not be practical for on-farm use (Molony and Kent, 1997). In contrast, using behavioral observations and visual assessments, caretakers of any skill can identify a management procedure or malady that is causing pain or discomfort. Behaviors such as tail flicking, stepping, kicking, shifting, and flinching have been used to indicate if a cow is experiencing pain or discomfort during milking (Willis, 1983, Van Reenan et al., 2002, Rousing et al., 2004, Rousing et al., 2006, Sutherland & Huddart, 2012, Cerqueira et al., 2017). The aim of this study was to investigate the prevalence and effects of UE on cow behavior in the milking parlor in first and second lactation Holstein dairy cattle.

MATERIALS AND METHODS

This study was approved by the Colorado State University's Animal Care and Use Committee (#1681). Cows in the first (FL) and second lactation (SL) that were between 2 and 9DIM were enrolled in the study.

Dairy Herds & Milking Procedure

Data were collected from 2 large (>500 head milking) Colorado dairies. Udder preparation procedures for Farm A and B are in Table 1.

Farm A

The milking parlor was a double, 20 herringbone parlor with a rapid exit. The milking claw vacuum system had a vacuum level of 45kPa. Farm A used the FutureCow brush (GEA Group Aktiengesellschaft, Düsseldorf, Germany) for washing and disinfecting the teats and drying the teats. The teat wash was a non-iodine, non-acid aqueous solution with chlorhexidine gluconate as the active disinfecting ingredient. A 1% iodine-based disinfectant was applied as a post-dip. Farm A had one pen of fresh cows ranging from 1 to 21DIM. All cows were fed ad libitum total mixed rations with 2 main

TABLE 1 – Udder Preparation Practices for Farm A and Farm B

	FARM A	FARM B
Step 1	Forestrip	Predip
Step 2	Wash and disinfect teat	Forestrip
Step 3	Dry teat	Wipe teat
Step 4	Attach milking unit	Attach milking unit
Step 5	Automatic take-off	Automatic take-off
Step 6	Post dip	Post dip

ingredients of corn silage and alfalfa silage. None of the first lactation cows had prior training through the milking parlor.

Farm B

The milking parlor was a double, 24 stall parallel parlor with a rapid exit. The milking claw vacuum system had a vacuum setting at 41 kPa. Farm B used the Thrifty Dipper T-150 Pro Complete Foamer System (Thrifty, Burley, Idaho) for application of the pre-dip. A 0.9% iodine post-dip was applied with a hand-held dip cup after milking was completed. Farm B had 2 pens of fresh cows ranging from 1 to 25DIM. Both fresh cow pens at Farm B were fed a similar total mixed ration. All rations included 2 main ingredients of corn silage and alfalfa silage. All cows were fed ad libitum total mixed rations as well as grazing pasture. None of the first lactation cows had prior training through the milking parlor.

Cows in Study

First or second lactation Holstein cows between 2 and 9DIM were enrolled in the study. Each farm was visited twice weekly and cows that were not previously observed were scored and recorded.

Video Set Up & Recording

Video recording began as cows entered the parlor. Each cow was visually identified by their ear tag by the first observer at the beginning of the milking procedure. Milking sessions were recorded with 8 GoPro cameras (GoPro Inc., San Mateo, CA). Each camera was fastened onto a secure spot to record cows from the rear while they were in the stalls of the milking parlor. Video recording of the milking session and scoring of UE occurred in the same milking session.

TABLE 2— Ethogram of Behaviors Collected in Study

Behavior	Definition
Step	Low, vertical leg lift (Hopster et al., 2002). When a cow shifted her rear legs, each raise of a foot off the floor was counted as a step.
Kick	Forceful and rapid movement of hind leg, often paired with direction of leg toward milker or milking unit (Van Reenen et al., 2002)
Kicking Off Milking Unit	A kick resulting in the forcible removal of the milking claw during the milking session.

Udder Edema Scoring

Udder edema was scored by the first author live at the end of milking. Per Morrison et al. (2018), a cow was indicated as having UE present when the appearance of the medial suspensory ligament was reduced and had uneven quarters, resulting in those with a score 1 or higher being converted to ‘edema present’. The presence of UE was assigned at the end of the milking session to avoid distortion of the udder from udder fill. The observer was standing in the milking pit and examined the cow in the milking stall from the rear to assign a UE score. The observer did not physically palpate any udders.

Behavior Scoring

Behaviors of stepping and kicking of the rear legs and kicking off the milking unit were documented and are defined in Table 2. All behavior data were recorded with 4 video observers. Each video observer received training to identify step and kick behaviors. Individual animals were randomly assigned to each video observer. To evaluate interrater reliability, raters were compared with a standard on the same video recording, with 12 sample videos. During evaluation, the number of steps during milking was considered, and bias and limits of agreement interval was calculated (Altman et al., 1983). The maximum estimated bias was 0.45 steps, and the widest limits of agreement interval was (−6.19, 5.89) for total steps during the milking period.

Observational time points of the milking procedure were broken down into the 9 categories defined in Table 3. When watching the recorded videos, observers recorded when a cow would step and kick when the milker was in contact or not in contact with the udder and when a cow would kick the milking unit off during the milking period. The entire milking session began at first contact of the udder during udder preparation. Observations during the active milking period were recorded for the first 5 min after complete attachment of the milking unit to avoid potential aversive behavior from over-milking. Any cows who completed their session before 5 min were noted and the total milking time was recorded. The separation of these behavioral ob-

servations matches the different phases of the milking procedure set by the Dairy Practices Council (udder preparation and unit attachment) along with matching the time points of udder preparation and active milking used by Kutzer et al. (2015).

STATISTICAL ANALYSIS

Data were analyzed using a linear regression model in version 4.1.1 of R (R Development Core Team, 2021). A sample size calculation identified that with a SD of 0.95, we could detect a difference of 0.3 kicks per minute with 80% power if we had 158 cows per lactation. Explanatory variables and covariates along with equation notation are listed in Table 3. During the udder preparation phase, the independent variables in the final model included UE score, lactation number, along with the covariates of DIM and the length of time the milker was in direct contact with the udder during prep.

$$\sqrt{(y)} = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4 + \beta_5X_5 + \beta_6X_1X_2 + \epsilon$$

Models for the attachment phase included the independent variables of UE score and lactation number, and the covariates of DIM, total prep time, and total time the milker was in contact with the udder during attachment of the milking unit.

$$\sqrt{(y)} = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4 + \beta_6X_6 + \beta_7X_1X_2 + \epsilon$$

The milking phase model included the independent variables of UE score and lactation number and the covariates of DIM, total prep time, and total milking time.

$$\sqrt{(y)} = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4 + \beta_7X_7 + \beta_8X_1X_2 + \epsilon$$

The interaction between UE score and lactation was examined in all models. Model assumptions for normality and equal variance were evaluated based on residual

TABLE 3 – Model Design for Analysis of Parlor Behavior in First and Second Lactation Cows with or without Edema

Explanatory Variables of Interest	Equation Notation	Description
Lactation number	β_1	First or second lactation
Edema Code - yes or no	β_2	Binary score for presence of udder edema
Covariates		
Days in Milk (DIM)	β_3	Number of days cow has been in current lactation
Duration of udder preparation in seconds	β_4	Total time elapsed for udder preparation phase
Duration milker was in contact with udder during prep	β_5	Total time elapsed during the udder preparation phase that the milker was in direct contact with udder
Duration milker was in contact with udder during attachment of milking unit	β_6	Total time elapsed during the milking unit attachment phase that the milker was in direct contact with udder
Duration of milking session	β_7	Total time elapsed for active milking period in seconds.

* Models analyzing data of rates during udder preparation with no contact were excluded from results as the rate were meaningful above zero.

plots. Our outcomes of interest, listed in Table 5, were converted into rate form. The relationship between each outcome and the explanatory variables of interest were examined by fitting a separate linear model for each outcome. When examining the rate of steps during udder preparation while the milker was in contact with the udder (**PICS**), a square root transformation was used in the model to address the equal variance assumption ($P = 0.077$). All other models did not use a transformation. All pairwise comparisons that were of interest and found to have $p\text{-value} \leq 0.05$ were analyzed using a Holm adjustment (Holm, 1979) for multiple comparisons.

RESULTS

Within the study population 237/247 (95.5%) of FL cows and 104/128 (81.3%) of SL cows presented with UE. Overall, 341/375 (90.7%) of cows within the study population presented with at least mild signs of UE; the softening of the medial cleft ligament. Distribution of the study population can be seen in Table 4.

All the cows with no edema stepped at least one time and 335/345 (98.5%) of cows with edema stepped at least one time. It was found that 190/236 (80.5%) of FL cows with UE kicked at least one time and 76/104 (73.1%) SL cows with UE kicked at least one time. Comparisons of interest are as follows: FL cows with UE compared with FL cows without UE, SL cows with UE compared with SL cows without UE, and FL cows with UE compared with SL cows without UE. Expected counts per session are presented in Table 6, Table 7, Table 8, and Table 9.

During the udder preparation phase when the milker was in contact with the udder (**PICS**), FL cows with UE had a higher rate of stepping at 3.24 steps/session as compared with FL cows without UE with 1.11 steps/session ($P = 0.049$). First lactation cows with UE had greater frequency of kick behavior at 2.68 kicks/session when compared with SL cows with UE during

the active milking session (**KM**) at 1.24 kicks/session ($P = 0.026$). First lactation cows with UE kicked off the milking unit (**KOU**) more often compared with SL cows with UE ($P = 0.005$). All pairwise comparisons were performed; practical comparisons with significance are shown in Table 10.

Various comparisons of behavior rates between treatment groups also bear biological significance. FL cows with UE had 2.80 steps/session and 1.45 kicks/session, FL cows without UE had 0.70 steps/session and 0.92 kicks/session, and SL cows with UE had 1.25 steps/session and 0.50 kicks/session during the attachment phase (**SA, KA**). SL cows with UE had 7.89 steps/milking session and 1.24 kicks/milking session, SL cows without UE had 8.35 steps/milking session and 0.83 kicks/milking session, FL cows with UE had 6.04 steps/milking session and 2.68 kicks/milking session, and FL cows without UE had 7.20 steps/milking session and 1.25 kicks/milking session (**SM, KM**). FL cows with UE had a higher average KOU count ($\mu = 0.220$ kick-offs) than FL cows without UE ($\mu = 0.091$ kick-offs) and SL cows with UE ($\mu = 0.029$ kick-offs).

DISCUSSION

Within this study, 90.7% of the sample population, ranging from 2 to 9DIM, was identified to have UE. Little industry-wide data about UE prevalence has been captured. The lack of this data may be due to

TABLE 4– Distribution of Population by DIM

DIM	First Lactation (n)	Second Lactation
2	18	1
3	57	12
4	46	22
5	51	40
6	35	27
7	24	19
8	16	6
9	0	1

TABLE 5 – Behaviors Recorded During Stages of the Milking Procedure

	Milker in contact or not in contact with the udder	Behavior	Abbreviations	Outcomes of Interest as Rates
During udder preparation	In Contact	Steps	PICS	Count of PICS divided by time in seconds milker was in contact with udder during preparation phase **
	No Contact	Steps	PNCS*	Count of PNCS divided by time in seconds milker was not in contact with udder during preparation phase
	In Contact	Kicks	PICK	Count of PICK divided by time in seconds milker was in contact with udder during preparation phase
	No Contact	Kicks	PNCK*	Count of PNCK divided by time in seconds milker was not in contact with udder during preparation phase
During milking unit attachment	NA	Steps	SA	Count of SA divided by time in seconds milker was in contact with udder during milking unit attachment phase
	NA	Kicks	KA	Count of KA divided by time in seconds milker was in contact with udder during milking unit attachment phase
During active milking	NA	Steps	SA	Count of SM divided by time in seconds milk unit was completely attached
	NA	Kicks	KA	Count of KM divided by time in seconds milk unit was completely attached
Kicking off the milking unit	NA	Kicks	KOU	NA

* Models analyzing data of rates during udder preparation with no contact were excluded from results as the rate were meaningful above zero.

** A square root of this was used to address model assumptions.

several reasons: UE being so common of a disorder that recording numbers on-farm does not warrant any benefit, caretakers are unaware of the negative impact UE has on various biological systems, or caretakers being unsure of how to manage UE. A binary scale was used for scoring UE in this study as it was a preliminary exploration of the relationship between UE and reactivity in the milking parlor. We acknowledge that a follow-up study which incorporates degrees of UE severity is warranted as the frequency of stepping and kicking behaviors may be affected by edema severity. Furthermore, investigating the impact UE has on parlor procedure timing could provide additional support for controlling the presence and severity of UE in dairy cows along with training employees to identify potentially reactive cows.

Fearful cattle are a threat to human safety and fear negatively affects productivity and welfare (Stenfelt, Yngvesson, and Rørvang, 2022). FL cows are more reactive, by moving their legs and kicking, when adapting to the milking routine (Hemsworth et al., 1989, Bremner, 1997). The research presented here was the first to investigate UE's effect on stepping and kicking behavior during the milking procedure. Results indicated that the presence of UE influenced the presentation of stepping and kicking behavior during various stages of the milking procedure. Low frequency rates (<0.02 behavior/time in seconds) of step and kick behavior during the udder preparation phase where the milker was not in contact with the udder (**PNCS**, **PNCK**) signify that reactivity primarily occurs during times of physically interacting with the edematous udder along

TABLE 6 – Edema Presence by Farm and Lactation Number

	Farm A		Farm B		Overall (n = 375)	
	First Lactation (n = 123)	Second Lactation (n = 74)	First Lactation (n = 124)	Second Lactation (n = 54)	First Lactation (n = 247)	Second Lactation (n = 128)
Edema Presence						
Edema Absent	8 (6.5%)	17 (23.0%)	3 (2.4%)	7 (13.0%)	11 (4.5%)	24 (18.8%)
Edema Present	115 (93.5%)	57 (77.0%)	121 (97.6%)	47 (87.0%)	236 (95.5%)	104 (81.3%)

TABLE 7— Summary of Behavior Types Presented Throughout Udder Prep and Active Milking in Total Population by Lactation and Presence of Edema

	First Lactation		Second Lactation		Overall	
	Edema Absent (no. of cows)	Edema Present (no. of cows)	Edema Absent (no. of cows)	Edema Present (no. of cows)	Edema Absent (no. of cows)	Edema Present (no. of cows)
Total Cows Per Group	11	236	24	104	35	340
Step Behavior						
No Step	0 (0%)	5 (2.1%)	0 (0%)	0 (0%)	0 (0%)	5 (1.5%)
Step	11 (100%)	231 (97.9%)	24 (100%)	104 (100%)	35 (100%)	335 (98.5%)
Kick Behavior						
No Kick	2 (18.2%)	46 (19.5%)	9 (37.5%)	28 (26.9%)	11 (31.4%)	74 (21.8%)
Kick	9 (81.8%)	190 (80.5%)	15 (62.5%)	76 (73.1%)	24 (68.6%)	266 (78.2%)
Kick Off Behavior						
No Kickoff	10 (90.9%)	195 (82.6%)	24 (100%)	101 (97.1%)	34 (97.1%)	296 (87.1%)
Kickoff	1 (9.1%)	41 (17.4%)	0 (0%)	3 (2.9%)	1 (2.9%)	44 (12.9%)
Number of Steps						
Mean (SD)	11.2 (14.4)	14.9 (11.8)	15.8 (10.5)	15.6 (11.8)	14.3 (11.8)	15.1 (11.8)
Number of Kicks						
Mean (SD)	3.73 (4.65)	6.20 (8.73)	2.67 (5.13)	3.05 (4.94)	3.00 (4.94)	5.24 (7.70)
Number of Kick-Offs						
Mean (SD)	0.09 (0.30)	0.22 (0.56)	0 (0)	0.03 (0.17)	0.03 (0.17)	0.16 (0.48)

with novel sensations. To separate reactivity based on novelty or UE, 2 populations of milking procedure-habituated FL cows, with and without udder edema, should be compared.

From our findings, it can be postulated that a cow with an edematous udder may experience discomfort during the milking session while their udder is handled. Diuretics have been shown to aid in the reduction of UE (Vestweber et al., 1989), but treatments of this na-

TABLE 8 - Rates of Behaviors During Observational Time points Converted to Number of Events During Time point for First Lactation Cows Categorized by Edema Presence

	Edema Absent 1st Lactation			Edema Present 1st Lactation		
	Behavior Count /Second Rate	Mean Duration	Behavior Counts per Session of Observational Time point	Behavior Count /Second Rate	Mean Duration	Behavior Counts per Session of Observational Time point
PICS	0.125 step/sec	8.91s	1.11 steps/session	0.321 step/sec	10.1s	3.24 steps/session
PICK	0.137 kick/sec	8.91s	1.22 kicks/session	0.169 kick/sec	10.1s	1.70 kicks/session
PNCS	0.011 step/sec	93.1s	1.02 steps/session	0.019 step/sec	137s	2.63 steps/session
PNCK	0.001 kick/sec	93.1s	0.09 kicks/session	0.0004 kick/sec	137s	0.05 kicks/session
SA	0.095 step/sec	7.45s	0.70 steps/session	0.275 step/s	10.2s	2.80 steps/session
KA	0.124 kick/sec	7.45s	0.92 kicks/session	0.143 kick/sec	10.2s	1.45 kicks/session
SM	0.025 step/sec	278s	7.20 steps/milking session	0.021 step/sec	280s	6.04 steps/milking session
KM	0.003 kick/sec	278s	1.25 kicks/milking session	0.009 kick/sec	280s	2.68 kicks/milking session

TABLE 9 – Rates of Behaviors During Observational Time points Converted to Number of Events During Time point for Second Lactation Cows Categorized by Edema Presence

	Edema Absent 2nd Lactation			Edema Present 2nd Lactation		
	Behavior Count /Second Rate	Mean Duration	Behavior Counts per Session of Observational Time point	Behavior Count /Second Rate	Mean Duration	Behavior Counts per Session of Observational Time point
PICS	0.273 step/sec	9.67s	2.63 steps/session	0.316 step/sec	9.30s	2.93 steps/session
PICK	0.131 kick/sec	9.67s	1.26 kicks/session	0.138 kick/sec	9.30s	1.28 kicks/session
PNCS	0.017 step/sec	117s	1.98 steps/session	0.021 step/sec	127s	2.66 steps/session
PNCK	0.001 kick/sec	117s	0.11 kicks/session	0.0006 kick/sec	127s	0.07 kicks/session
SA	0.271 step/sec	5.71s	1.54 steps/session	0.270 step/s	4.65s	1.25 steps/session
KA	0.085 kick/sec	5.71s	0.48 kicks/session	0.108 kick/sec	4.65s	0.50 kicks/session
SM	0.029 step/sec	288s	8.35 steps/milking session	0.028 step/sec	282s	7.89 steps/milking session
KM	0.002 kick/sec	288s	0.83 kicks/milking session	0.004 kick/sec	282s	1.24 kicks/milking session

ture would be costly and laborious. Therefore, it is recommended that caretakers approach the management of UE through preventative practices. Okkema and Grandin (2021) review the identified risk factors of UE, including that the etiology of the disorder is currently unknown. Uncovering the root cause of udder edema would provide a targeted approach for prevention and treatment of the disorder. With our current knowledge of the disorder, next steps should focus on delivering an economical and practical therapy to reduce the negative effects of UE to aid producers in mitigating this disorder, both in the short and long-term.

Within the data, we anticipated finding FL cows with UE to have greater stepping and kicking rates than FL cows without edema and all second lactation cows, with or without edema. Interestingly, we found that SL cows with UE had a greater step rate during PICS than FL cows without UE. It may be worthwhile to investigate the effect of total udder preparation time on the step and kick behavior in cows when examining parlor behavior in association with UE. By adjusting for this, observations can evaluate if cows with UE have ideal milk letdown when compared with cows without UE and if edematous udders require longer udder preparation times for adequate milk letdown.

CONCLUSIONS

Udder edema was prevalent (90.7%) and has a negative impact on parlor behavior during the udder preparation period when the milker is in contact with the udder and during milking. It is paramount to conduct further research to better understand the etiology of UE to reduce incidence-rate and severity and improve animal wellbeing and productivity. An industry-wide epidemiological survey of UE to quantify nationwide prevalence can identify a new, widespread challenge for the dairy industry to address and solve. Dissemination of the currently known risk factors and negative ef-

fects of UE along with finding practical and economical methods of treatment and prevention of UE would aid managers in reducing the prevalence of UE in their herd. Further research is warranted to fully understand the relationship between stepping and kicking behavior during the milking procedure in cows with UE. The data from this study provide insight on how UE in dairy cattle affects both parlor efficiency and animal welfare.

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TABLE 10 – Behaviors with Statistically Significant Rates Between Lactation and Udder Edema Presence

Behavior	Contrasts between Lactation Number and Edema Presence*			
	Contrast	Estimated Difference in Kick-Off Count	p-value, $\alpha = 0.05^{\dagger}$	Cow Group with Higher Rate
PICS	1 No – 1 Yes**	–0.213 ¹	0.049	1 Yes
KM	1 Yes – 2 Yes	0.005 ²	0.026	1 Yes
KOU	1 Yes – 2 Yes	0.172	0.005	1 Yes

*All pairwise comparisons were performed but those of practical application are shown here.

** 1 stands for first lactation, 2 stands for second lactation, No stands for edema absent, Yes stands for edema present.

[†] p-values presented are with Holm adjustment for multiple comparisons.

¹a negative estimated difference in behavior rate indicates that the first animal category has a lower behavior rate than the second animal category.

²a positive estimated difference in behavior rate indicates that the first animal category has a higher behavior rate than the second animal category.

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